BIRESWAR SETT

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SUMMARY

Computer Science and Data Science undergraduate focused on building systems and tools from first principles, spanning kernel-level programming, GPU/CUDA computation, and applied machine learning. Experienced in translating ambiguous, real-world problems into working research and engineering outcomes - from a from-scratch operating system kernel to a professor-validated algorithmic heuristic targeting IEEE publication. Comfortable owning a problem end-to-end, working across abstraction layers, and communicating technical work clearly to both engineers and researchers.

EDUCATION

Indian Institute of Technology, Madras — *Chennai, Tamil Nadu*
Bachelors in Data Science and Applications — *Jul. 2024 - Sept 2028*

Siksha 'O' Anusandhan University — *Bhubaneshwar, Odisha*
B.Tech in Computer Science and Engineering — *Aug. 2024 - Aug 2028*

PROJECTS

Akasha | Link — Aug 2026 – Sept 2026

- Led development of an agentic vision-language platform for natural language analysis of optical, SAR and multi-temporal remote sensing imagery.
- Designed the workflow for query understanding, task routing, invoking specialist model/tool and evidence-grounded response generation over heterogeneous geo-spatial inputs.
- Implemented and deployed production facing infrastructure for model serving, API/backend services, cloud deployment, authentication, storage, and containerized execution.
- Led a team of six engineering student across various domains to design, interpret and debug the project turning a real life ambiguous problem statement of ISRO to a deployment ready system.

K-centre algorithm — Feb 2026 - Present

- Developing a structural heuristic for 1D k-centre optimization that exploits point distribution, anchor selection, gap structure, and recursive cluster splitting to reduce the effective search space.
- Implemented and evaluated successive algorithmic variants against exact solutions and Gonzales' heuristic across 500 edge cases, with the current approach recovering 212 exact optima in O(n log n).
- Deriving structural conditions on optimal centre configurations for the k<n-1 and formalizing a lemma that constrains when optimal centres can be adjacent.

StarOS - Physics Simulation OS | Link — Oct 2025 - Present

- Built a custom operating system from scratch, implementing the bootloader and transition from 16-bit real mode to 32-bit protected mode.
- Designing a modular kernel architecture intended to support a custom scheduler for real-time N-body physics simulations.
- Developing low-level system components including memory management, interrupt handling, and hardware initialization.

Smart MCQ Solver | Link — July 2026 - Aug 2026

- Developed a custom Transformer encoder with shared embeddings and cosine-similarity based ranking, tailored for multiple-choice question answering.
- Benchmarked against BERT and RoBERTa baselines, performing hyperparameter tuning and experiment tracking using Weights & Biases.
- Evaluated performance using Mean Average Precision (MAP) with custom transformer encoder model scoring best at ~0.75 on public dataset on Kaggle.

TECHNICAL SKILLS

Languages: C++, C, Python, Java, C#, JavaScript, SQL (PostgreSQL, SQLite), HTML/CSS, x86 Assembly, Bash
Frameworks: Flask, Django
Developer Tools: Git, GitHub, Vim, VS Code, Visual Studio, Linux
Libraries: PyTorch, Scikit-learn, Hugging Face Transformers, NumPy, Pandas, Matplotlib, CUDA